Examination Statistical Methods

Linköpings Universitet, IDA, Statistik

Course code and name:	732A83 Statistical Methods
Date:	2025/05/13, 8–12
Examinator:	Krzysztof Bartoszek
Allowed aids:	Pocket calculator
	Table with common formulæ and moment generating functions (distributed with the exam)
	Table of integrals (distributed with the exam)
	Tables with distributions, probabilities and percentage points (distributed with the exam)
	One double sided A4 page with own hand written notes
Grades:	A= $[18-\infty)$ points
	B= $[16-18)$ points
	C= $[14-16)$ points
	D= $[12-14)$ points
	E= $[10-12)$ points
	F= $[0-10)$ points
Instructions:	Write clear and concise answers to the questions.
	Reporting on the exam a probability outside the $[0,1]$ interval or a negative variance **without a comment** that a mistake must have been made somewhere will result in a **NEGATIVE** point for each such instance.

Problem 1 (2p)

You are in a class of 4 students which gets randomly divided into two groups of four (i.e., each student has the same chance of getting into the same group.

1. (1p) What is the probability that your best friend (you have only one) will be in the same group as you?
2. (1p) What is the probability that your best friend (you have only one) will not be in the same group as you?

Problem 2 (2p)

Consider two events A and B such that $P(A) = 0.5$, $P(A \setminus B) = 0.3$ and $P(A \cup B) = 0.7$

1. (1p) What is the probability that event B takes place?
2. (1p) What is the probability that event A took place if we know that B took place?

Problem 3 (3p)

Consider a discrete random variable $X \in \{0, 1, 2, 3\}$ with probability distribution $P(\{0\}) = 0.2$, $P(\{1\}) = 0.1x$, $P(\{2\}) = 0.2x$, $P(\{3\}) = 0.1x$.

1. (1p) Calculate x.
2. (1p) Left $F_X(\cdot)$ be the cumulative distribution function of X. Find $F_X(-100)$, $F_X(2)$, $F_X(3)$ and $F_X(100)$.
3. (1p) Find the expectation and variance of X.

Problem 4 (3p)

A random variable X has has density function

$$f_X(x) = \begin{cases} 4x^c & 0 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

1. (2p) Calculate c.
2. (1p) Calculate the mean and variance of X.

Problem 5 (3p)

Assume that you have an i.i.d. sample of n elements from the Gaussian $\mathcal{N}(\mu, 1)$ distribution, and the variance, equalling 1 is known. Construct, with proof, a confidence interval, at the α level, for μ.

Problem 6 (4p)

Consider the following game. You start off with 0kr and toss a fair coin and every time it shows heads you win 1kr, if tails, then you win 0kr. Let S_n be the total winnings up to toss n.

1. (2p) What is the distribution of S_n?
2. (2p) Calculate the mean and variance of S_n.

Problem 7 (3p)

Consider a Poisson process, $N(t)$, with intensity 0.5.

1. (1p) What is the expected number of events on the time interval $[0, 1]$?
2. (1p) What is the probability that $P(N(2) = 1)$?
3. (1p) What is the expected time between two jumps of the process?